

## INSPECTION CERTIFICATE

Date: 01/10/2018

FILE NO. 18.15610-1

**COMPLIANCE**  
With IEC 61508:2010

**SGS Italia Spa**  
hereby confirm that

**OFFICINE OROBICHE S.P.A**  
Via SERENA, 10  
24010 PONTERANICA (BG) - ITALY

**LEVEL TRANSMITTERS**  
2000 /TLT

Has been assessed per the requirements of:  
IEC 61508:2010 parts 1-7

And meets requirements providing a level of integrity to:

**SYSTEMATIC INTEGRITY : SC 2 - SIL 2 CAPABLE**  
**RANDOM INTEGRITY: TYPE A ELEMENT**  
SIL 2 @HFT=0; ROUTE 1<sub>H</sub>  
SIL 2 @HFT=0; ROUTE 2<sub>H</sub>

**MEET THE SIL REQUIREMENTS DETAILED IN THE ANNEXED TABLE**  
**CAN BE USED UP TO SIL2 FOR THE SAFETY FUNCTION:**  
"correct switching on demand"



This document is issued by the Company under its General Conditions of Service accessible at <http://www.sgs.com/en/Terms-and-Conditions.aspx>. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein.

Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and punishable by law.

**SGS Italia S.p.A.**  
Via Caldera, 21  
20153 Milano – Italy  
Industrial

Inspection Certificate  
18.15160-1  
October 01 2018  
Pg 1 di 2

## FOR PRODUCTS TYPE A HIGH DEMAND MODE

### SUMMARY TABLE

|                                                  |                                               |
|--------------------------------------------------|-----------------------------------------------|
| E/E/PE Safety related system<br>(sensor element) | Officine Orobiche Spa                         |
| Family product                                   | <b>Level indicators/ transmitters (LIT)</b>   |
| Safety Function definition                       | Correct operation of measure and transmission |
| MAX SIL allowable                                | <b>SIL2</b>                                   |
| HFT                                              | 0(100%)                                       |
| PFH [h <sup>-1</sup> ] – high demand mode        | 4,76E-08 h <sup>-1</sup>                      |
| SFF                                              | 94%                                           |
| DC                                               | 62%                                           |
| $\lambda_{DU}$                                   | 48 FIT                                        |
| $\lambda_{DD}$                                   | 76 FIT                                        |
| $\lambda_{SD}$                                   | 571 FIT                                       |
| $\lambda_{SU}$                                   | 86 FIT                                        |
| MTTR                                             | 8h                                            |
| Useful Lifetime                                  | 10 years                                      |
| Proof Test suggested                             | 12 months                                     |

A safety Instrument Function (SIF) designed with this product must not be used at a SIL level higher than stated without "prior in use" justification by end user or diverse technology redundancy in the design.

Technical report: 15.500.0375 Rev.01 dtd. 01/06/2017

Surveillance report: 18.500.0886 dtd. 27/09/2018

Reference standard: IEC 61508-1/7:2010

Expiry date : 30/07/2021

**SGS Italia S.p.A.**

On behalf of Business Manager  
Andrea Zani



This certificate emendates and substitutes the previous 8475/MI rev.1 issued 01/06/2017  
This certificate is composed of 2 pages



SGS



This document is issued by the Company under its General Conditions of Service accessible at <http://www.sgs.com/en/Terms-and-Conditions.aspx>. Attention is drawn to the limitation of liability indemnification and jurisdiction issues defined therein.

Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders will be prosecuted to the fullest extent of the law.

SGS Italia S.p.A.  
Via Caldera, 21  
20153 Milano – Italy  
Industrial

Inspection Certificate  
18 15160-1  
October 01 2018  
Pg 1 di 2